Asphalt Pavement Pothole Detection using Deep learning method based on YOLO Neural Network.

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***Abstract*— There is an increasing need for assessment of national road condition. Currently, some automatic devices have been extensively applied to collect up-date data about road condition, such as the use of survey vehicle for collecting data— which make it faster and more accessible, and semi-automatically data processing that is useful for policy decision making. Yet, demand for more detail road data is continuously growing; thus, data improvement needs to perform, upgrading the existing solution. To date, stages on identification and classification of road damages are being conducted semi-manually based on images collected by survey vehicle; it is hindered due to the facts that this method is the cost-consuming process and may result in inconsistency. Therefore, this present work used YOLO with three different architecture configuration, i.e., Yolo v3, Yolo v3 Tiny, and Yolo v3 SPP, enabling us to create a more accurate assessment for detecting potholes on the road surface. The results showed the average mAP values for Yolo v3, Yolo v3 Tiny, and Yolo v3 SPP at 83.43%, 79.33%, and 88.93%. While the area measurement shows the accuracy of 64.45%, 53.26%, and 72.10% respectively. And it needs 0,04 second to detect each image. Conclusively, it shows a satisfying result in pothole detection; thus, this technique has a high opportunity to developed and implemented as a tool for road assessment.**

Keywords— Pothole Detection, YOLO, Computer Vision, Object Detection, Distress Detection.

I. INTRODUCTION.

Road construction is one of the government's priorities. Therefore, Every year targeted to increase national roads length. These raises also improving road maintenance and road rehabilitation programs. Consequently, it is crucial to monitoring the road condition to determine the proper programs.

Monitoring road conditions or referred to as road conditions assessment consists of three stages, i.e., data collection, identification & classification, and the last is the road conditions assessment [1]. In developing countries [2] and especially in Indonesia, data collection and road distress identification can be done manually with a survey form filled in by surveyors or using a specific survey vehicle to take road surfaces pictures followed by road distress identification using image processing software. Both of these methods require high costs, both in terms of time and labor [3][4] and inconsistencies in the assessment process, because it depends on the surveyor or operator experience [5]. With lots of segments that must be inspected regularly, the automation process is essential to do. On the other hand, the regulations pursue more detailed road condition data. As in *Penentuan Indeks Kondisi Perkerasan* (IKP) [6], there are twenty types of distress to assessed. This condition needs for continuous improvement to complete the existing methods, so the road conditions appropriately presented. From several distress types, Pothole is one of an important indicator of road defects [7]. Therefore, it needs proper detection to determine the appropriate road maintenance program.

In the pothole detection method, various studies have been carried out with different approaches, for example using a laser [8], vibration sensor [9], [10] and image-based detection, both two-dimensional [11] and three-dimensional using a stereo camera [12][13]. The development of advanced image processing technique, easy image interpretation, and the availability of low-cost camera acquisition devices have inspired the development of image-based pothole detection. Likewise, with the presence of new methods in the computer vision study, especially related to objects detection, give new opportunities to implement it in the road damage detection. In this study, YOLO (You Only Look Once), as one of the state-of-the-art methods in the field of object detection, is chosen as a pothole detection method. This research, hopefully, provides an alternative pothole solution on the road surface for road distress identification and classification as a complement to existing methods.

This paper is written in the following order, section I contains the introduction, section II is related research and literature studies, section III is the proposed method, section IV is our result experiment, and discussion and the last is the research conclusion.

II. RELATED WORKS AND LITERATURE.

A. Related Works.

Koch et al. [4] define three visual specific characteristics of the pothole, i.e., oval in shape, dark in color and rougher surface than its surrounding. Form these three characteristics, shape extraction, image segmentation, and feature extraction is compared to its surroundings to determine a specific value that represents a pothole. Complementing and improving the previous method, Koch et al. [1] proposed an enhanced pothole detection technique to incrementally updates the previous method. They used video as a base of their research material. Therefore, in the newly proposed method, they utilize tracking and counting pothole in pavement video. Moreover, it has cut off processing time to inspect pothole in every frame.

L. Huidrom et al. [14] proposed a method to classify three types of road distress, i.e., pothole, crack, and patch with distress area measurement. They proposed two-step detection methods, classify frame with distress using the DFS algorithm. Then followed by assign distress category and area measurement using CDDMC algorithm. They used image texture, shape factor, and dimension as the key feature to assign distress category.

In addition to image processing-based detection, technological developments in machine learning also encourage pothole detection more adaptively. Nhat-Duc Hoang [2] proposed an artificial intelligence (AI) model for

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detecting pothole on the asphalt pavement surface. They use image processing method as a tool for extracting the critical feature of a pothole. They are Gaussian filter, steerable filter, and integral projection. Then, assign to pothole class using two machine learning algorithms, i.e., the least-squares support vector machine (LS-SVM) and the artificial neural network (ANN).

The previous approaches have been beneficial in the pothole detection process, but have disadvantages where feature extraction must be performed by experts to produce excellent accuracy performance during the detection process [15]. This leads the deep learning development in the computer vision study. Deep learning able to compute features extraction and classification simultaneously through convolutional neural network operations.

H. Maeda et al. [16] proposed a distress detection method with eight distress type at once. Focusing on provide dataset accessible by the public, they capture the road surface using a low-cost smartphone in several municipal in Japan. So they provide 9.053 images with 15.435 distress. In a detection process, they compare SSD MobileNet dan Inception V2 performance. They provide a considerable contribution where inspection and analyzation of road distress possible to perform in a mobile application. However, this study has not covered comprehensive area calculations.

In this study, we focus on processing road distress image data found in developing and tropical countries, that is Indonesia. We also carry out extensive pothole area measurement with raw data acquired from a specific survey vehicle, which the data retrieval process is done using a rear camera facing downwards so that the detection results can be equipped with a measurement area.

B. YOLO Algorithm.

Deep learning, also known as a deep neural network, is a method used to process raw data and automatically find the representation of features needed to do classification or detection [15]. To determining representatives features, deep learning uses the neural network in their operations [17]. Deep learning is usually composed of many layers and uses convolution operations to find characteristic features known as convolutional neural networks (CNN). CNN raises new developments in computer vision, especially detection objects. Detection object is the process of classifying and determining the object location at once [18].

There are two frameworks used for object detection, i.e., based on region proposals and based on regression or classification operations. Included in the proposal base region category are SPP-Net [19], R-CNN Variant [20]–[23], and FPN [24]. While in regression/classification based framework category are SSD [25] and YOLO [26]. YOLO, especially YOLO v3 in object detection provides accuracy approaching SSD, but in terms of detection time, YOLO v3 provide detection time the times faster than SSD [27]. Therefore, this study uses YOLO v3 as an architecture to do pothole detection.

YOLO uses a single step to predict the whole image, classification, and localization at once. The whole image features are used to predict classes and bounding boxes simultaneously. Image input will be divided into S x S grid, called the grid cell. Each grid cell predicts B bounding box by containing five pieces of information, i.e., the center of the object (x, y), the length and width of the object (w, h) and the value of confidence score and C class probabilities [26]. So the final target prediction is denoted as $S \times S \times (B \times 5 + C)$ tensor.

In this study, we use three architectures of Yolo v3, i.e., Yolo v3, Yolo v3-tiny, and Yolo v3-spp. Yolo v3 is an original Yolo v3 with 53 layers feature extractor, then Yolo v3 Tiny is a small version of Yolo v3 while Yolo v3 is Yolo v3 combine with Spatial Pyramid Pooling (SPP) [28].

III. Proposed Method.

The phase taken in this study is the preparation of research data, followed by annotation and labeling. The prepared data is divided into training data and testing data. The labeled train data used to build a model using a Yolo v3 architecture. The output of the modeling phase is a model or also called the weight. To assess the performance of the model, we do a detection and area measurement with the testing data. The proposed method is shown in Fig. 1.

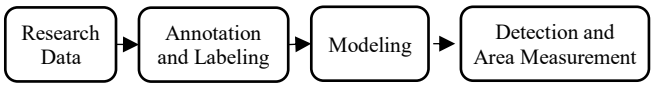


Fig. 1. Proposed Method.

A. Research Data.

We use highway image data with a pothole in it, which was taken in 2016 from eight segments groups in East Java province. The Pavement View camera took the image on the Hawkeye 2000 survey vehicle. A red box in Fig.2 shows the Pavement View camera installation illustration.

The pavement view camera creates black and white videos with .AVI format. This video will be processed using the Hawkeye processing toolkit software. Then exported to a .jpg file format. Image results can be selected with various sizes, i.e. 1280x960, 1624x1234, 2048x1536 and 512x384 pixels which represents 2,061x1,391 meters in real conditions.



Fig. 2. Hawkeye survey vehicle with the pavement view camera [29].

Exporting data offers many images according to the road segment length. Therefore, it needs to do a visual assessment to get an image with a pothole in it. Finally, we select 448 images and divided into train and test data. The detail of data distribution between train and test data is 50%: 50%, 60%: 40%, 70%: 30%, 78%: 22%, 80%: 20% and 90%: 10%. Among the earlier test with the next test has 10% of data decreasing. So, there are 224 train data and 224, 150, 96, 64, 56, and 25 test data, respectively. While the potholes data distribution has a similar percentage with image data, which is 49%: 51%, 58%: 42%, 69%: 31%, 79%: 21%, 81%: 19%,

and 91%: 9%. The detail distribution of train and test data are shown Table I.

TABLE I. DATA DISTRIBUTION.

No.	Train Data		Test Data		Total	
	Images Number	Potholes Number	Images Number	Potholes Number	Images Number	Potholes Number
1	224	279	224	296	448	575
2	224	279	150	199	374	478
3	224	279	96	124	320	403
4	224	279	64	73	288	352
5	224	279	56	64	280	343
6	224	279	25	27	249	306

B. Annotation and labeling

Annotation and labeling is the process of marking potholes in images. The way is to create a bounding box in the pothole outer part and give it a label. This process is assisted by label tools to produce YOLO framework input formats. Annotation and labeling were carried out on all research data. The information stored in the annotation and labeling process are id_class, bounding box center (x and y), and bounding box width and height (w and h). The id_class is an integer value, starting from 0 and the bounding box information is decimal format at a 0-1 scale. Each image in the .jpg format will have a file in .txt format with pothole information. Examples of annotation and labeling are shown in Fig 3.

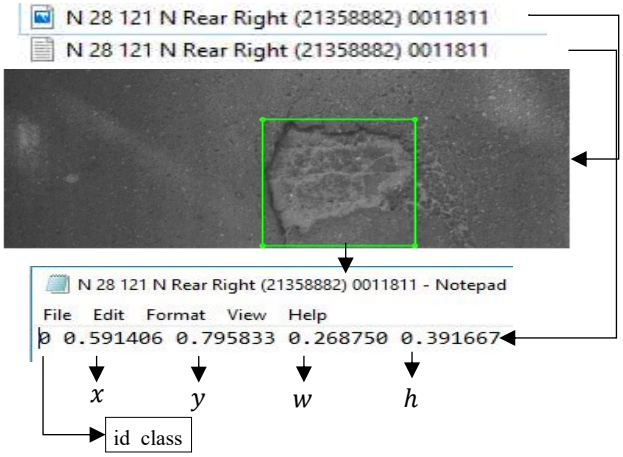


Fig. 3. Annotation and Labeling.

C. Modeling.

The pothole modeling process is carried out using three architectures of YOLOv3, i.e., YOLOv3, YOLOv3 tiny, and YOLOv3-SPP with a pre-trained model based on Imagenet [30]. The pothole modeling process was built using a darknet framework with C and CUDA languages, which runs through Google Collaboratory [31] based on Jupyter notebook with computer specifications of Intel (R) Xeon (R) CPU @ 2.30GHz, 13 Gb Memory, GPU Tesla T4 and Ubuntu 18.04.1 operating system.

The pothole modeling process is done in 10000 iterations using three architectures of Yolo v3, i.e., Yolo v3, Yolo v3 Tiny, and Yolov3 SPP with 224 train data. The parameter setting is 0,001 learning rate configuration, $416 \times 416 \times 3$ input configuration, 64 batch size with 16 subdivisions, and $13 \times 13 \times 18$ output target configuration. We save a model in every hundred iterations that will evaluate to get the performance of the model.

D. Detection and Area Measurement.

In the detection process, all model is tested using six different number of test data, i.e., 224, 150, 96, 64, 56, and 26 test data. The purpose of this test is to obtain performance value by comparing the detection result with the ground truth. We use IoU (1), precision (2), recall (3), and mAP as a parameter evaluation.

$$IoU = |A \cap B| / |A \cup B| \quad (1)$$

Intersection over Union (IoU) is the overlap area of a ground truth bounding box (A) and the detection bounding box (B) divided by the union are of both bounding boxes. IoU represents how similar a detection object with ground truth object.

$$Precision = TP/TP + FP \quad (2)$$

$$Recall = TP/TP + FN \quad (3)$$

We use a ratio between *TP, FP*, and *FN* in precision and recall. True Positive (*TP*) is the correct detection; pothole object detected as a pothole. False Positive (*FP*) is the false detection, not a pothole object but detected as a pothole. Whereas False Negative (*FN*) is an object should be a pothole but detected as not a pothole. While, mean Average Precision known as mAP is a single number represents a combination of values between precision and recall, also known as Under Curve Area (AUC) [32]. The IoU used as the mAP parameter is 0.5.

While area measurement evaluation is carried out to obtain Accuracy of measurement. Accuracy (Acc) is achieved by comparing the ground truth area and the detection area. Where we define the ground truth area as L and the detection area as L' so we get a measurement accuracy with equations (4).

$$Acc = 100 - \frac{|L-L'|}{L} \times 100\% \quad (4)$$

Where L is width and height multiplication, which width is w and height is h, by the range of 0-1. So, it needs coefficients multiplier to get the actual pothole area. The coefficient from the Hawkeye processing toolkit software shows that each frame of an image represents 2,061 × 1,391 meters in actual size so that the area of L can be obtained according to the equation (5).

$$L = w \times h \times 2{,}061 \times 1{,}391\, m^2 \quad (5)$$

IV. RESULT AND DISCUSSION.

A. Modeling result.

In the pothole modeling process, the loss is an important result. Each modeling process produces its loss. So we can combine the three loss to see different performance of the modeling process as shown in Fig.4. We can see that Yolo v3-tiny produces a higher loss compared to the other two architectures. While Yolov3 and Yolo v3 SPP provide almost the same loss. At the beginning iteration, the loss for the three architectures is quite significant, but after 2000 iterations the loss is smaller than one and get constant fluctuations. Overall loss shows convergent values towards 0 with the increasing iterations number. Each Yolo v3, Yolo v3 Tiny and Yolo v3 SPP architectures up to 10,000 iterations generate a minimum loss of 0.01, 0.04 and 0.01. While at the end of the iteration, each generates losses of 0.02, 0.06, and 0.018. We also evaluate the loss generated in the last 100 iterations indicating

that the change value is quite small, which is an average of 0.0049, 0.0147, and 0.0049.

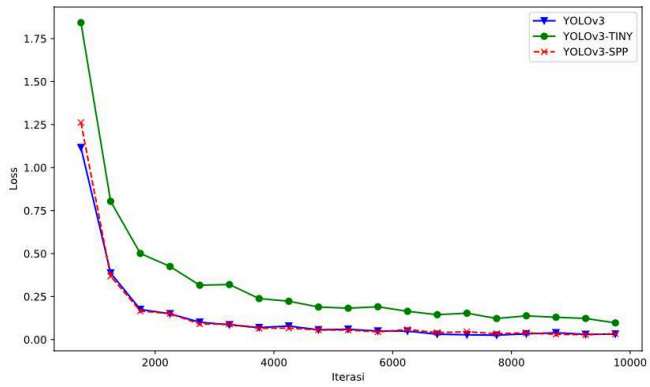


Fig. 4. Loss of Yolo v3, Yolo v3 Tiny and Yolo v3 SPP in the modeling process.

B. Evaluation Accuracy.

The results of evaluations to get accuracy from each model are shown in Table II. From our experiment, Yolo v3 delivers average precision, recall, IoU, and mAP of 0.92, 0.75, 71.60%, and 83.43%, Yolo v3 Tiny deliver average precision, recall, IoU and mAP of 0.94, 0.64, 71.07% and 79.33%, While Yolo v3 SPP deliver average precision, recall, IoU and mAP of 0.94, 0.82, 72.59% and 88.93%.

TABLE II. DETECTION ACCURACY.

Arch	Test Data	TP	FP	FN	Prec	Rec	IoU (%)	mAP (%)
Yolo v3	224	220	28	76	0.89	0.74	68.43	81.00
	150	147	21	52	0.88	0.74	68.09	78.77
	96	87	8	37	0.92	0.70	72.29	80.46
	64	54	4	19	0.93	0.74	74.04	86.90
	56	52	1	12	0.98	0.81	75.03	86.92
	25	20	2	7	0.91	0.74	71.69	86.52
	Avg.				0.92	0,75	71,60	83.43
Yolo v3 Tiny	224	182	23	114	0.89	0.61	67.21	76.16
	150	121	13	78	0.90	0.61	69.03	75.38
	96	74	5	50	0.94	0.60	71.69	75.55
	64	48	2	25	0.96	0.66	73.41	82.89
	56	42	1	22	0.98	0.66	75.14	83.41
	25	18	1	9	0.95	0.67	69.94	82.59
	Avg.				0.94	0,64	71,07	79.33
Yolo v3 SPP	224	244	21	52	0.92	0.82	70.12	86.93
	150	155	16	44	0.91	0.78	69.27	84.32
	96	101	8	23	0.93	0.81	70.75	85.75
	64	60	3	13	0.95	0.82	73.32	91.02
	56	52	3	12	0.95	0.81	73.89	90.28
	25	23	1	4	0.96	0.85	78.21	95.27
	Avg.				0.94	0,82	72,59	88.93

The object detection accuracy is usually noticed from the mAP value. So, it is essential to increasing precision value over recall value to get better mAP value by increasing the TP ratio. From Table II, we can see that Yolo v3 SPP architecture generate higher TP number and lower FP number than the other two architectures. With increasing of twenty-four TP numbers and decreasing of seven FP numbers than Yolo v3 made, so it is certainly presenting a better precision and mAP. We can conclude Spatial Pyramid Pooling added in the Yolo v3 architecture increasing mAP by 5,5%, and it makes Yolo v3 SPP provide the best mAP in our experiment. While the IoU value from the three architecture does not show a significant difference.

C. Area Measurement Accuracy.

The best mAP from evaluation accuracy is used to get the detection area and calculate them to get area measurement accuracy. We get area measurement accuracy by comparing the total of detection area with the ground truth area from all images. For example, if we have detection at two images with a pothole at each image, where $w'_1 = 0.009$, $h'_1 = 0{,}158$, $w'_2 = 0{,}323$, and $h'_2 = 0{,}329$. While the ground truth is $w_1 = 0{,}276, h_1 = 0{,}387, w_2 = 0{,}079,$ and $h_2 = 0{,}103$, then the area measurement accuracy can be shown in the following calculation:

$$L_1 = w \times h \times 2{,}061 \times 1{,}391\ m^2$$
$$L_1 = 0.079 \times 0.103 \times 2.061 \times 1.391\ m^2 = 0.023\ m^2$$
$$L_2 = 0.306\ m^2$$
$$L'_1 = w' \times h' \times 2{,}061 \times 1{,}391\ m^2$$
$$L'_1 = 0.099 \times 0.158 \times 2.061 \times 1.391\ m^2 = 0.045\ m^2$$
$$L'_2 = 0.304\ m^2$$
$$Acc = 100 - \left|L - L'\right|/L \times 100\%$$
$$Acc = 100 - \frac{|(0.023 + 0.306) - (0.045 + 0.304)|}{(0.023 + 0.306)} \times 100\ \%$$
$$Acc = 100 - \frac{0.019}{0.330} \times 100\% = 100 - 0.058 \times 100\%$$
$$Acc = 100 - 5.8\ \% = 94.17\ \%$$

We apply that calculation to all test images to get total area measurement accuracy. Each Yolo v3, Yolo v3 Tiny, and Yolo v3 SPP architectures deliver average area measurement accuracy of 64.45%, 53.26%, and 72.10%, in more detail, shown in Table III. During our experiment, although the mAP value presents a quite good result in detection, it does not apply to area measurement. While if we compared with the IoU, the area measurement accuracy also remains a smaller value. Comparison of mAP, Area measurement accuracy, and IoU are shown in Fig. 5.

TABLE III. AREA MEASUREMENT ACCURACY.

Arch	Test Data	Ground Truth Area (m^2)	Prediction Area (m^2)	Deviation		Accuracy (%)
				(m^2)	(%)	
Yolo v3	224	48.53	29.51	19.03	39.20	60.80
	150	30.43	21.40	9.03	29.67	70.33
	96	17.98	11.85	6.13	34.09	65.91
	64	9.74	6.47	3.27	33.61	66.39
	56	8.77	5.83	2.94	33.52	66.48
	25	5.57	2.95	2.62	47.04	52.96
	Total	121.02	78.00	43.02	35.55	64.45
Yolo v3 Tiny	224	48.53	26.73	21.80	44.93	55.07
	150	30.43	16.81	13.62	44.76	55.24
	96	17.98	9.38	8.60	47.82	52.18
	64	9.74	4.59	5.15	52.87	47.13
	56	8.77	4.02	4.76	54.21	45.79
	25	5.57	2.93	2.64	47.38	52.62
	Total	121.02	64.45	56.56	46.74	53.26
Yolo v3 SPP	224	48.53	36.45	12.08	24.89	75.11
	150	30.43	21.47	8.96	29.45	70.55
	96	17.98	12.81	5.16	28.71	71.29
	64	9.74	6.75	2.98	30.65	69.35
	56	8.77	6.21	2.56	29.22	70.78
	25	5.57	3.55	2.01	36.19	63.81
	Total	121.02	87.25	33.77	27.90	72.10

The lower area measurement accuracy occurs because of the total area comparison between the detection ($TP + FP$) area compared to ground truth ($TP + FN$) area. So if the model fails to detect objects in an image (FN), later, it will

decreasing area measurement accuracy. While of the overall tests, the average percentage of FN is 26% of all the ground-truth objects. Then, it will decrease the area measurement accuracy by 26 % indirectly. Otherwise, if we only selected images with the detected objects in it, even though success (TP) or not (FP) with the selected ground truth images, then, it will increase the area measurement accuracy by 26 %. So, it will give measurement accuracy over 78 % even doing the simplest model of Yolo v3 Tiny. While for Yolo v3 architecture gives a measurement accuracy of 90.45% and Yolo v3 SPP of 98.10%.

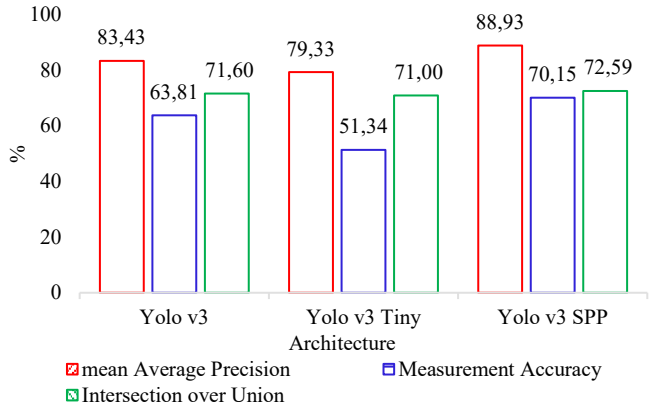


Fig. 5. Mean Average Precision, Area Measurement Accuracy, and Intersection over Union.

Nevertheless, we do not use the selected image with the detected object in it to calculate the area measurement accuracy. We instead choose a total comparison of all detection and all ground truth images. It is based on the images number generated in a road segment is very large. So, it is helpless if we have to check frame by frame to get the presence of pothole, then do an area measurement. Also, the assumption in the distress area measurement, distress area recapitulation in a road segment is more important than the distress area of frame by frame. It is implied that focused on reducing the FN ratio for further research could increase the detections number and the measurement accuracy.

D. Visual Assesment.

We do a detection from the best mAP and presenting some result in Fig 6. From all of the image, we select some false negatives and some true positives image with the best confidence to analyze. In Fig. 6. we presenting a ground truth as a white color bounding box, while a detection bounding box as the other color. Fig 6.a, 6b, 6c, 6d, and 6e are false detection, while Fig 6f, 6g, and 6h are the true positive.

Fig. 6a is a false detection because the pothole object is quite small and mixed with cracks. While Fig. 6b and 6c are false detections because the damage position is on the side of the image. It is often happening even though our training data has pothole at the side of the image. The detection failed in Fig. 6d because of ambiguous pothole shape, While Fig. 6e failed because of the pothole size is quite large while there is no much data train which a large size shown in Fig. 6e. And, also overlapping with a small-sized object. While for true positive in Fig 6f, 6g, and 6h, it can be seen, almost all the detection bounding boxes have difference size compared to the ground truth bounding box. It contributes to making an error in the area measurement accuracy.

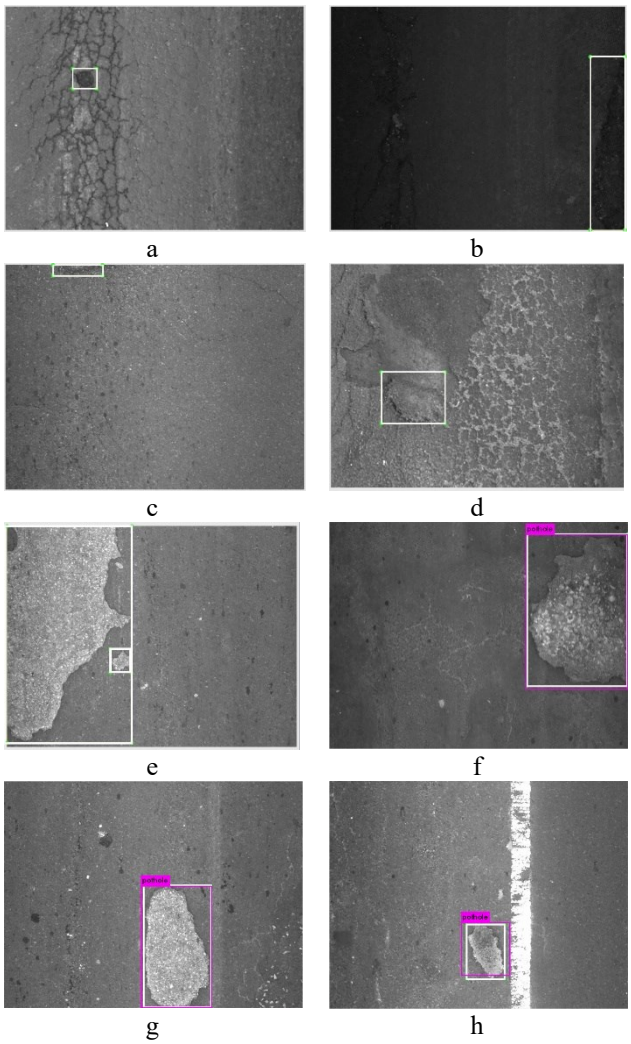


Fig. 6. Visual detection result. a, b, c, d, e : False detection. f, g, h: Success detection.

E. Detection Time.

We also carry out a detection time assessment with three different architectures, as shown in Table IV. Our experiment shows that Yolo v3 Tiny provides the fastest detection time; it is in line with the mAP accuracy in our evaluation accuracy, where there is a trade-off between accuracy and detection time. Moreover, Yolo v3 Tiny CNN complexity architecture also more straightforward than the others. It will be considered in pothole detection implementation, in case time is a priority, Yolo v3 Tiny is the best, but if accuracy is a priority, Yolo v3 SPP is the right one. Overall the detection time shows the fairly good speed of 0.04 second per image.

TABLE IV. DETECTION TIME PER IMAGE

Test Data	Yolo v3 (s)	Yolo v3 Tiny (s)	Yolo v3 SPP(s)
224	0.0402	0.0357	0.0402
150	0.0400	0.0400	0.0400
96	0.0417	0.0313	0.0521
64	0.0469	0.0313	0.0313
56	0.0357	0.0357	0.0536
25	0.0400	0.0400	0.0400
Avg.	0.0407	0.0357	0.0428

V. CONCLUSION.

During our experiment, it can be concluded that our model with Yolo neural network succeeded in detecting potholes for asphalt pavement image. It shows a satisfactory detection

accuracy of our applied architecture, Yolo v3, Yolo v3 tiny and Yolo v3 SPP. Which present the mAP of 83.43%, 79.33%, and 88.93% respectively, and the area measurement accuracy of 64.45%, 53.26%, and 72.10% respectively. And it needs average detection time in 0.04 second per image. Therefore, it has a high opportunity to developed and implemented.

Although the accuracy and detection time in pothole detection is satisfying, further research and involvement of civil engineers need to be considered to get better class definitions. So, it will provide more distress class and benefits for the civil engineer in analyzing road condition, especially in Indonesia.

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